Ruiqi **Ding**

Research Assistant · Freshly Graduated (22 years old)

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Gradinaru Lab

August 27, 2020

California Institute of Technology
1200 E California Blvd
Pasadena, CA 91125

Application for PhD study

Dear Dr. Gradinaru,

About Me

My GPA (3.616/4) is not perfect, and I haven't achieved the best results in every course, but I have made great efforts and got outstanding results in my two favorite fields of quantum physics and biophysics. For example, during the undergraduate exchange at the University of Göttingen (Riemann, von Neumann, von Karman...), I chose all master biophysics courses and achieved excellent results. At the National University of Singapore, my alma mater, I would audit doctoral-level courses whenever I had time.

Although I don't have the best talent, I still rely on my interest and passion to spend almost all of my spare time volunteering to experience scientific research in the professors' labs, never minding that they don't contribute to my transcript. Through self-study and exploration in these projects, I have mastered engineering and programming skills that are far beyond the scope of standard curriculum framework of physics.

PhD research interest

Light is the particle that physicists are most interested in, and research on it has led to numerous Nobel Prizes. From the first x-rays to the recent LIGO and optical tweezers, this makes me believe even more that optogenetics is the most promising technology in the field of neuroscience. I hope to participate in research that overcomes the existing difficulties of optogenetics, such as penetration and attenuation problems. While auditing my professor Liu Xiaogang's nanochemistry class, I learned that his team used up-conversion nanoparticles to convert near-infrared light outside the skull into blue light near nerve cells, achieving non-invasive deep brain stimulation. I believe that my physical background will help this kind of exploration based on physical principles.

Sincerely,

Ruiqi Ding

Attached: Curriculum Vitae